

Vittorio Zampinetti

Research Scientist | Applied Machine Learning

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Summary

Five years of applied ML research — probabilistic generative models, variational inference, and large-scale experiment pipelines. Published at RECOMB 2024. Strong in Python, PyTorch, and JAX, with mathematical foundations in probability theory and optimisation.

Experience

PhD in Applied Mathematics and Computational Biology

Nov 2021 - Sep 2026

Politecnico di Torino and KTH Royal Institute of Technology

- Designed and published VICTree (RECOMB 2024) — a variational inference system for Bayesian tumor phylogeny reconstruction; ELBO optimisation over tree-structured latent variables using PyTorch and JAX
- Built large-scale experiment infrastructure on HPC clusters: end-to-end training, systematic evaluation, and comparative benchmarking across model variants
- Developed probabilistic data models for high-dimensional, noisy single-cell sequencing inputs — preprocessing, feature extraction, and quality validation at scale
- Maintained a shared research codebase used by collaborators at two institutions throughout a five-year program

Visiting Researcher

May 2023 - Sep 2023

Memorial Sloan Kettering Cancer Center, New York

- Built ML processing components for whole-genome sequencing workflows in a clinical research setting.
- Delivered structured, validated outputs for downstream statistical modelling.

Research Engineer

Sep 2021 - Feb 2022

KTH Royal Institute of Technology, Stockholm

- Developed and validated multi-layer probabilistic sequence models (HMMs) for biological pattern recognition in Python.
- Delivered clean, documented implementations for long-term use by the research group.

Teaching Assistant, R for Statistics (MSc)

Oct 2022 - Sep 2024

Politecnico di Torino

- Designed and delivered MSc-level lectures on statistical programming and reproducible analytical workflows in R.

Education

PhD, Applied Mathematics & Computational Biology – Politecnico di Torino & KTH, Expected Sep 2026

MSc, Data Science and Engineering - Politecnico di Torino, Graduated with Honors

BSc, Computer Engineering - Universita di Firenze, Graduated with Honors

Awards

Alta Scuola Politecnica — excellence programme for selected students of Politecnico di Torino and Politecnico di Milano, 2019–2021. asp-poli.it

Skills

Languages: Python, C/C++, R

ML & Research: PyTorch, JAX, NumPy, Pyro, TensorFlow, variational inference, probabilistic modeling

Infrastructure: HPC/SLURM, Docker, Git, Unix, Claude Code

Publications

- VICTree - A Variational Inference Method for Clonal Tree Reconstruction, RECOMB 2024
- Sampling random spanning arborescence in graphs with low conductance, Statistics and Probability Letters, accepted 2025
- Cellmates: a method for single-cell phylogeny reconstruction with proper evolutionary distances from copy numbers, bioRxiv 2026
- Scalable and robust phylogenetic tree reconstruction from copy number data with Sparse Rooted Neighbor Joining, bioRxiv 2026